Skills Review

Exercise No. 16

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Course/Section: CPE4B **Date: Oct. 7, 2026**

Machine Problem:

Write a program that will create an output similar below and asks the user to input the necessary information. Use the functions 06h, 02h, 0h, 0Ah, and 07h or 01h. Save the program using Exer16.asm as the filename.

Program:

```
; AARON LUDWIG A. ALTAR
; CPE4B

.model small
.stack
.data

form    DB '              +-------------------------------------------------+',13,10
        DB '              |       University of Batangas - Lipa Campus      |',13,10
        DB '              |               Lipa City, Batangas               |',13,10
        DB '              |                                                 |',13,10
        DB '              |         Computer Engineering Department         |',13,10
        DB '              |                  Student Record                 |',13,10
        DB '              |                                                 |',13,10
        DB '              |     Student Name       :                        |',13,10
        DB '              |     Student Number     :                        |',13,10
        DB '              |     Year Level         :                        |',13,10
        DB '              |     Gender             :                        |',13,10
        DB '              |     Birthdate          :                        |',13,10
        DB '              |     Age                :                        |',13,10
        DB '              |                                                 |',13,10
        DB '              |                                                 |',13,10
        DB '              |       Programmed by: Aaron Ludwig A. Altar      |',13,10
        DB '              +-------------------------------------------------+',13,10,'$'

again   DB 'Input another record? [Y/N]: $'

snpar   label byte            ; parameter list for student name
snlen   DB 24                 ; maximum number of characters
actsnlen DB ?                 ; actual length entered
snfld   DB 24 dup(' ')        ; characters entered

nopar   label byte            ; parameter list for student number
nolen   DB 24
actnolen DB ?
nofld   DB 24 dup(' ')

ylpar   label byte            ; parameter list for year level
```

```
yllen   DB 24
actyllen DB ?
ylfld   DB 24 dup(' ')

gdpar   label byte              ; parameter list for gender
gdlen   DB 24
actgdlen DB ?
gdfld   DB 24 dup(' ')

bdpar   label byte              ; parameter list for birthdate
bdlen   DB 24
actbdlen DB ?
bdfld   DB 24 dup(' ')

agpar   label byte              ; parameter list for age
aglen   DB 4                    ; up to 3 digits + Enter
actaglen DB ?
agfld   DB 4 dup(' ')

.code
start:
        mov ax, @data           ; initialize the data segment
        mov ds, ax

restart:
        ;** clear the screen **
        mov ah, 06h             ; Function 06h - scroll up / clear window
        mov al, 00h             ; AL = 0 - clear the whole window
        mov bh, 07h             ; attribute - white on black
        mov cx, 0000h           ; upper left corner (row 0, col 0)
        mov dx, 184Fh           ; lower right corner (row 24, col 79)
        int 10h

        ;** display the student record form **
        mov ah, 02h             ; Function 02h - set cursor position
        mov bh, 00h             ; page 0
        mov dx, 0200h           ; row 2, col 0
        int 10h

        mov ah, 09h             ; Function 09h - display string
        mov dx, offset form     ; whole bordered form
        int 21h

        ;** student name **
        mov ah, 02h
        mov bh, 00h
        mov dx, 0929h           ; row 9, col 41 (after the colon)
        int 10h

        mov ah, 0Ah             ; Function 0Ah - buffered string input
        mov dx, offset snpar
        int 21h
```

```
;** student number **
mov ah, 02h
mov bh, 00h
mov dx, 0A29h          ; row 10, col 41
int 10h

mov ah, 0Ah
mov dx, offset nopar
int 21h

;** year level **
mov ah, 02h
mov bh, 00h
mov dx, 0B29h          ; row 11, col 41
int 10h

mov ah, 0Ah
mov dx, offset ylpar
int 21h

;** gender **
mov ah, 02h
mov bh, 00h
mov dx, 0C29h          ; row 12, col 41
int 10h

mov ah, 0Ah
mov dx, offset gdpar
int 21h

;** birthdate **
mov ah, 02h
mov bh, 00h
mov dx, 0D29h          ; row 13, col 41
int 10h

mov ah, 0Ah
mov dx, offset bdpar
int 21h

;** age **
mov ah, 02h
mov bh, 00h
mov dx, 0E29h          ; row 14, col 41
int 10h

mov ah, 0Ah
mov dx, offset agpar
int 21h

;** ask to input another record **
```

```
        mov ah, 02h
        mov bh, 00h
        mov dx, 1419h          ; row 20, col 25
        int 10h

        mov ah, 09h
        mov dx, offset again   ; "Input another record? [Y/N]:"
        int 21h

getkey: mov ah, 07h            ; Function 07h - read a key without echo
        int 21h                ; AL = key pressed

        cmp al, 'Y'            ; is it Y?
        je yes                 ; if so, new record
        cmp al, 'y'            ; also accept lowercase y
        je yes
        cmp al, 'N'            ; is it N?
        je finish              ; if so, end the program
        cmp al, 'n'            ; also accept lowercase n
        je finish
        jmp getkey             ; any other key - wait for Y or N

yes:    jmp restart            ; restart is too far for je, so use jmp

finish:
        mov ah, 02h            ; move the cursor below the form
        mov bh, 00h            ; so the DOS prompt will not overlap
        mov dx, 1600h          ; row 22, col 0
        int 10h

        mov ah, 4Ch            ; terminate program
        int 21h

end start
```

Output:

```
+-------------------------------------------------+
|       University of Batangas - Lipa Campus      |
|               Lipa City, Batangas               |
|                                                  |
|         Computer Engineering Department          |
|                  Student Record                  |
|                                                  |
|     Student Name        : Aaron Ludwig A. Altar  |
|     Student Number      : 2320616                |
|     Year Level          : 4th Year               |
|     Gender              : Male                   |
|     Birthdate           : July, 18. 2004         |
|     Age                 : 22                     |
|                                                  |
|                                                  |
|       Programmed by: Aaron Ludwig A. Altar       |
+-------------------------------------------------+
```